

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

STRUCTURE OF ATOM

LECTURE-2

CHEMISTRY

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Topics To Covered

- 1** **Discovery of Proton**
- 2** **Discovery of Neutron**
- 3** **Thompson Model of Atom**

REVISION



DISCOVERY OF ELECTRON →

- By J. J. Thompson
- Discharge tube experiment
- PROPERTIES OF CATHODE RAYS.

- Material particle
- Heating effect
- -vely charged.
- Ionisation effect.

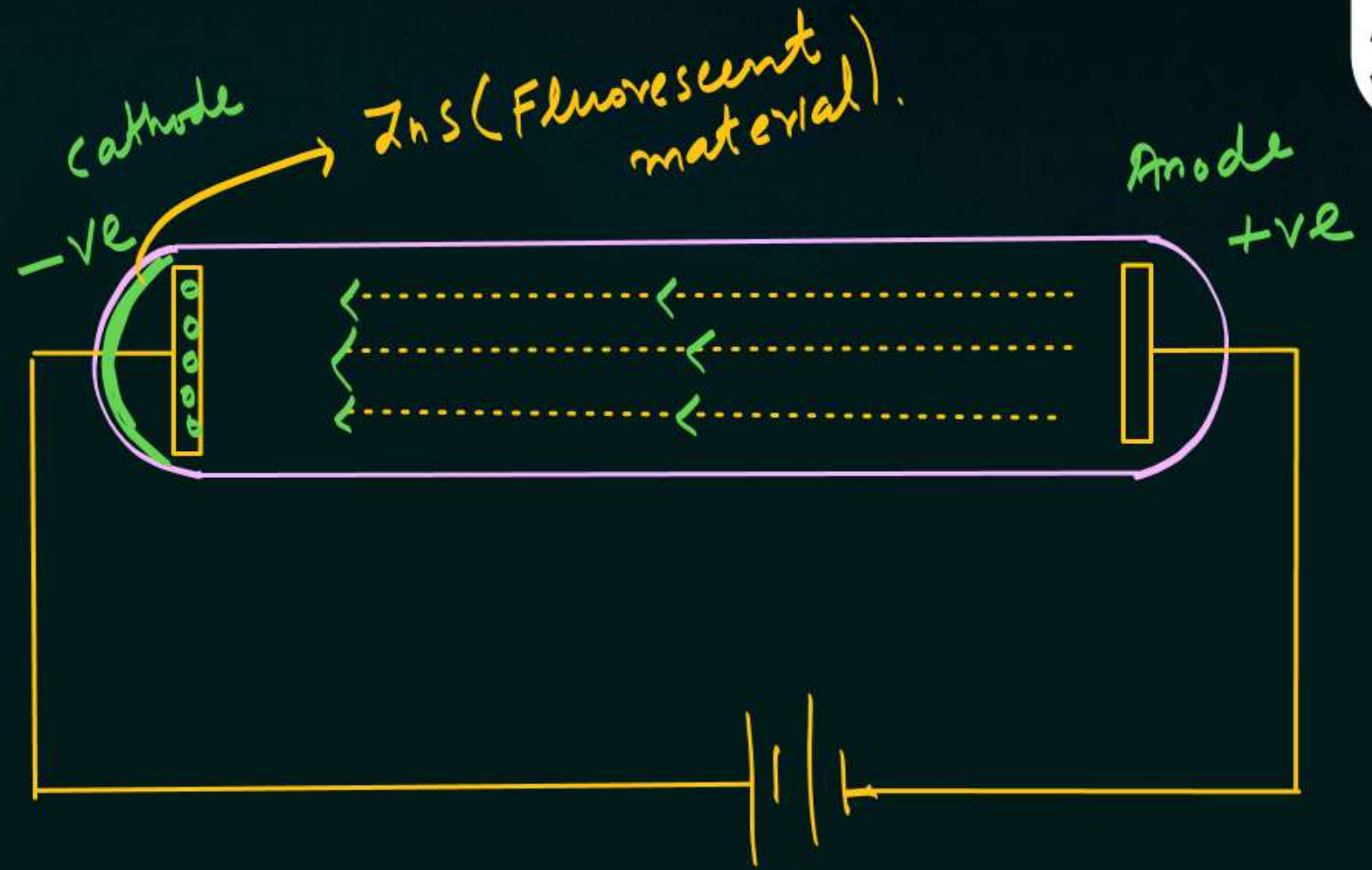
$$\frac{e}{m_e} = 1.73 \times 10^{11} \text{ C/kg.}$$

$$e = 1.6 \times 10^{-19} \text{ C } (-1).$$

$$m = 9.1 \times 10^{-31} \text{ kg.}$$

Discovery of Proton

- In 1836, E. Goldstein discovered that in addition to cathode rays, a new kind of rays are also found streaming behind the cathode in discharge tube experiments. These rays travelled in opposite direction to the cathode rays.
- This mean that these rays consist of positively charged particles and were also named positive rays or canal rays or anode rays.



Discovery of Proton



Some of the characteristics of anode rays are:

1. The anode rays travel in straight lines and cast shadow of the object placed in their path.
2. The anode rays are deflected by the magnetic and electric fields like cathode rays. But the deflection of anode rays is in the opposite direction to that of the cathode rays. For example, these rays are attracted towards the negative plate in the electric field. This means that these rays consist of positively charged particles.
3. Like cathode rays, these rays also rotate the wheel placed in their path and also have heating effect.
4. The charge to mass ratio (e/m) for these rays is considerably smaller than electrons.
5. Unlike cathode rays, the e/m ratio of positive rays depends upon the nature of the gas taken in the tube.

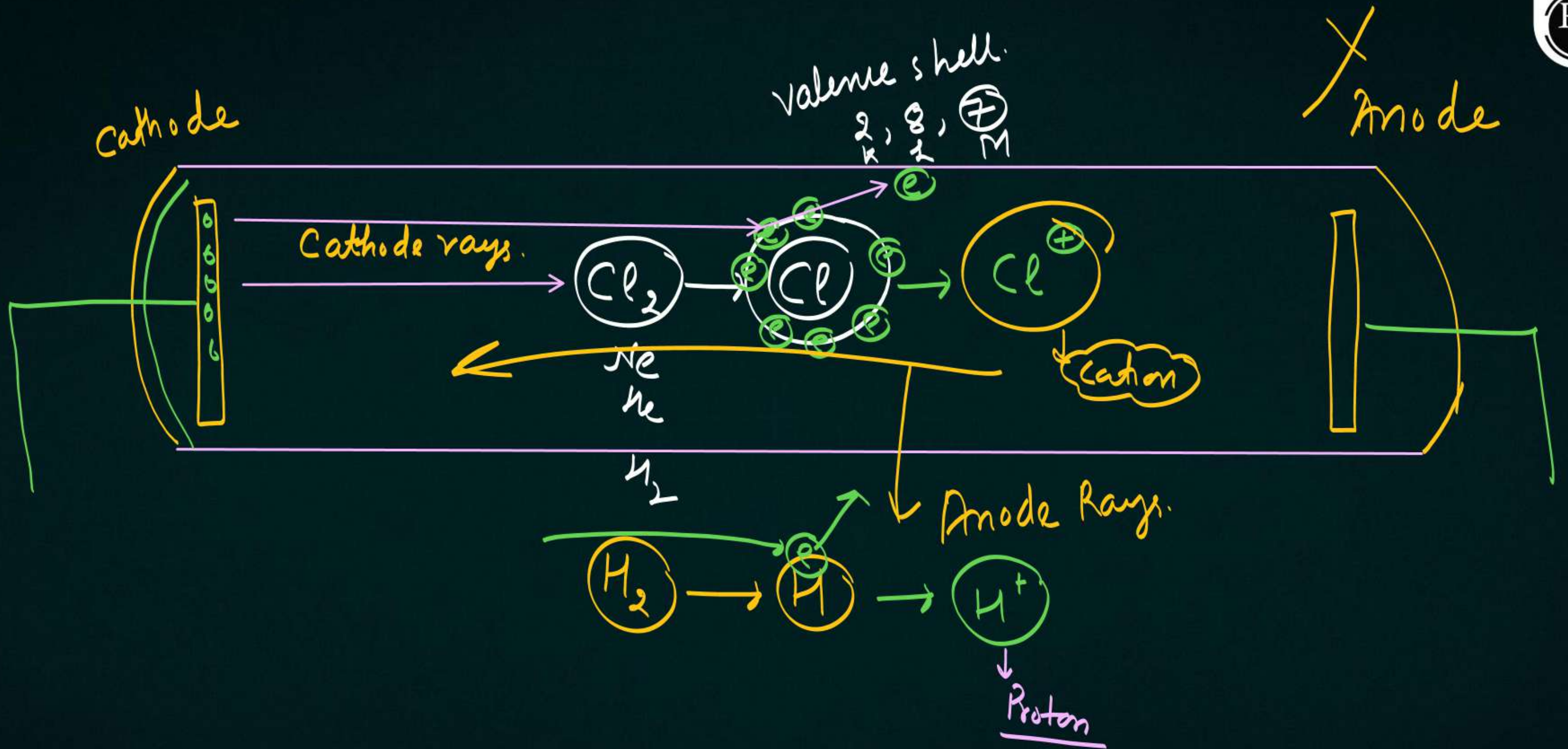
consist of material particle.

$$\left(\frac{e}{m}\right)_p < \left(\frac{e}{m}\right)_e$$

ORIGIN OF CANAL RAYS.

- Canal rays do not originate from the surface of Anode.
- Canal rays originate from the space present between two metal electrodes.
- Cathode rays strike the gas present in discharge & convert it into cation & these cations move from anode to Cathode are called as Anode rays.





CHARGE TO MASS RATIO OF PROTON: →

Charge to mass ratio for proton is calculated when Hydrogen gas is placed discharge tube. It was found to be

$$\frac{e}{m} = 9.58 \times 10^7 \text{ C/kg}$$

Charge of proton was found be $= 1.6 \times 10^{-19} \text{ C}$

OR Relative charge of proton = +1.

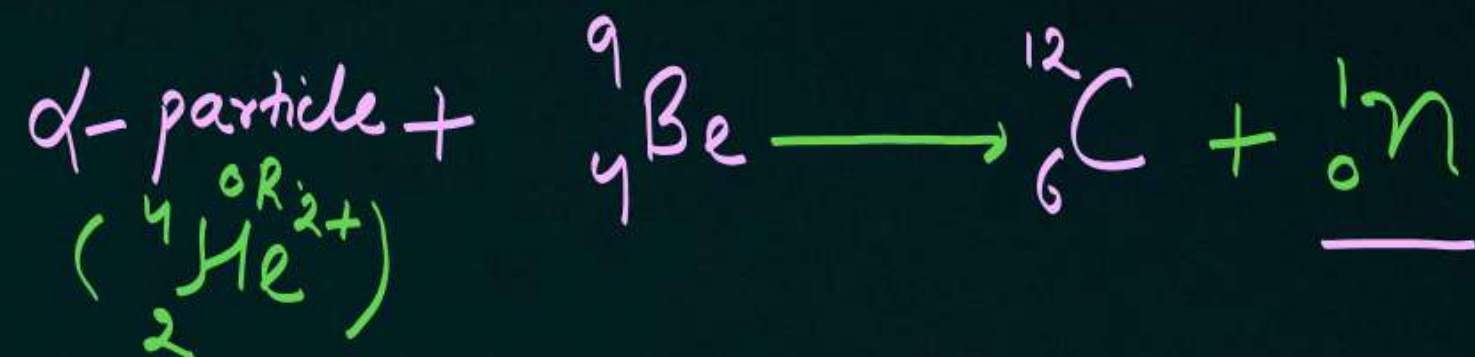
$$\begin{aligned} \text{Therefore the mass of proton (m)} &= \frac{e}{e/m} \\ &= \frac{1.6 \times 10^{-19} \text{ C}}{9.58 \times 10^7 \text{ C/kg}} \\ &= 1.67 \times 10^{-27} \text{ kg} \end{aligned}$$

• Therefore proton can be defined as "sub atomic particle having one unit positive charge ($1.6 \times 10^{-19} \text{ C}$) & mass equal to the hydrogen atom ($1.67 \times 10^{-27} \text{ kg}$).

• Electron can be defined as "sub-atomic particle having one unit -ve charge ($1.6 \times 10^{-19} \text{ C}$) and mass equal to $\frac{1}{1837}$ of Hydrogen atom. ($9.1 \times 10^{-31} \text{ kg}$).

Discovery of Neutron

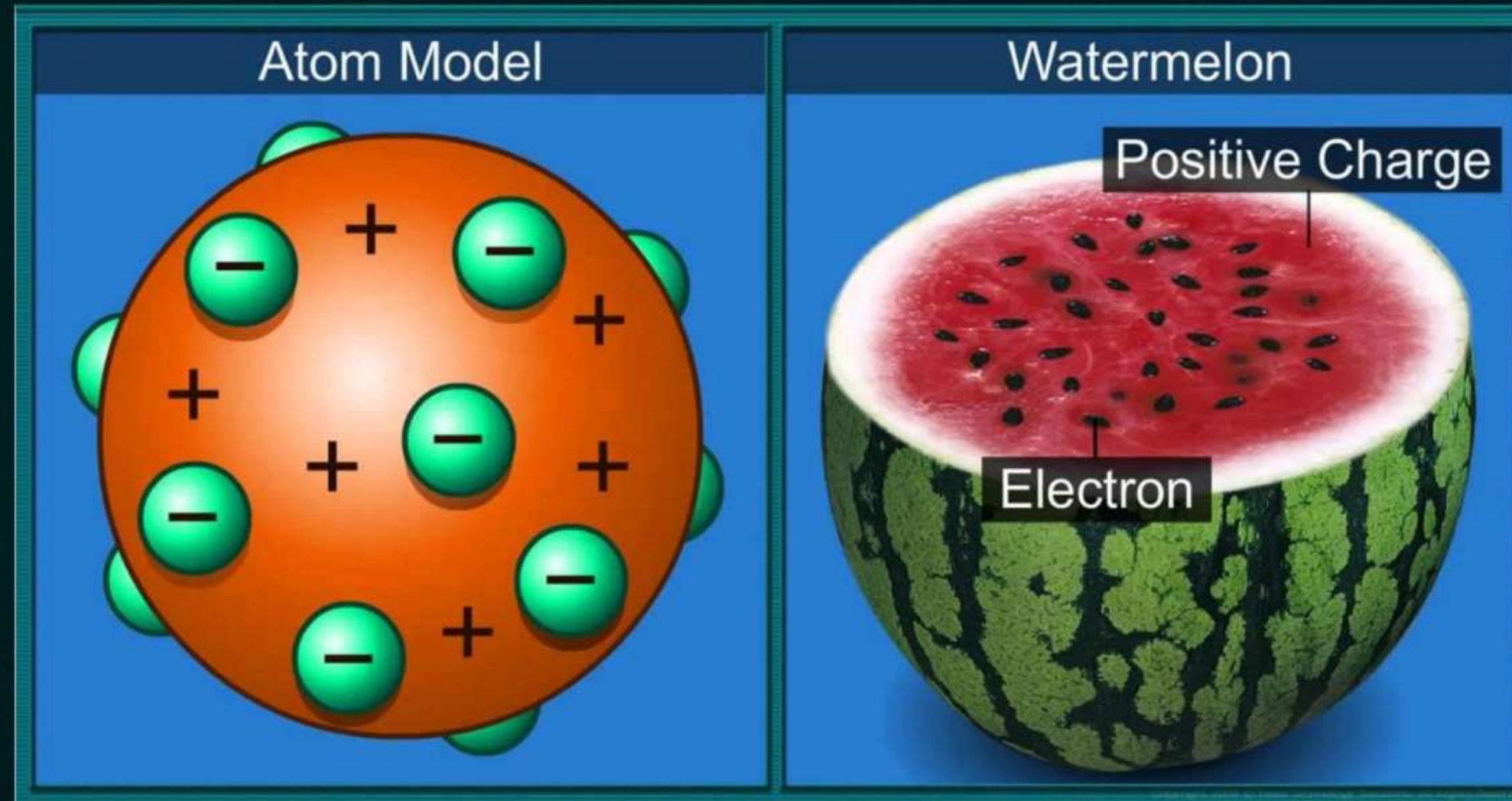
- Neutron was discovered by James Chadwick with the help of scattering experiment. In this experiment α -particles were bombarded on the thin sheet of Beryllium. A neutral particle have mass equal to the mass of proton were emitted. It was named as neutron.



Therefore neutron is defined as, sub-atomic particle having no charge & mass equal to the hydrogen atom.

	Electron	PROTON	NEUTRON
CHARGE	$1.67 \times 10^{-19} \text{ C } (-1).$	$1.67 \times 10^{-19} \text{ C } (+1).$	ZERO
e/m	$1.75 \times 10^{11} \text{ C/kg.}$	$9.58 \times 10^7 \text{ C/kg.}$	ZERO
m	$9.1 \times 10^{-31} \text{ kg}$	$1.67 \times 10^{-27} \text{ kg.}$	$1.67 \times 10^{-27} \text{ kg.}$

THOMSON'S ATOMIC MODEL



QUESTION



Which of the following particles has the smallest mass?

1

Electron

2

Proton

3

Neutron

4

Alpha particle

QUESTION



What is the relative charge of a neutron?

1 +1

2 -1

3 0

4 +2

QUESTION

If the relative charge of an electron is -1 , what is the absolute charge in coulombs?

1

$1.6 \times 10^{-19} \text{ C}$

2

$-1.6 \times 10^{-19} \text{ C}$

3

$3.2 \times 10^{-19} \text{ C}$

4

$-2.2 \times 10^{-19} \text{ C}$

QUESTION



What is the ratio of charge to mass for a proton if the charge is 1.602×10^{-19} C and mass is 1.672×10^{-27} kg?

1

$$9.58 \times 10^7 \text{ C kg}^{-1}$$

2

$$1.758 \times 10^{11} \text{ C kg}^{-1}$$

3

$$9.58 \times 10^{10} \text{ C kg}^{-1}$$

4

$$1.76 \times 10^8 \text{ C kg}^{-1}$$

QUESTION



Which of the following particles was the first to be discovered using the modified cathode ray tube?

1 Neutron

2 Proton

3 Electron

4 Photon

QUESTION



Who discovered the neutron by bombarding beryllium with alpha particles?

1 J.J. Thomson

2 Millikan

3 Rutherford

4 Chadwick

QUESTION



Calculate the number of electrons which will together weigh one gram.

Thank
You